

Smart Glasses

For blind people

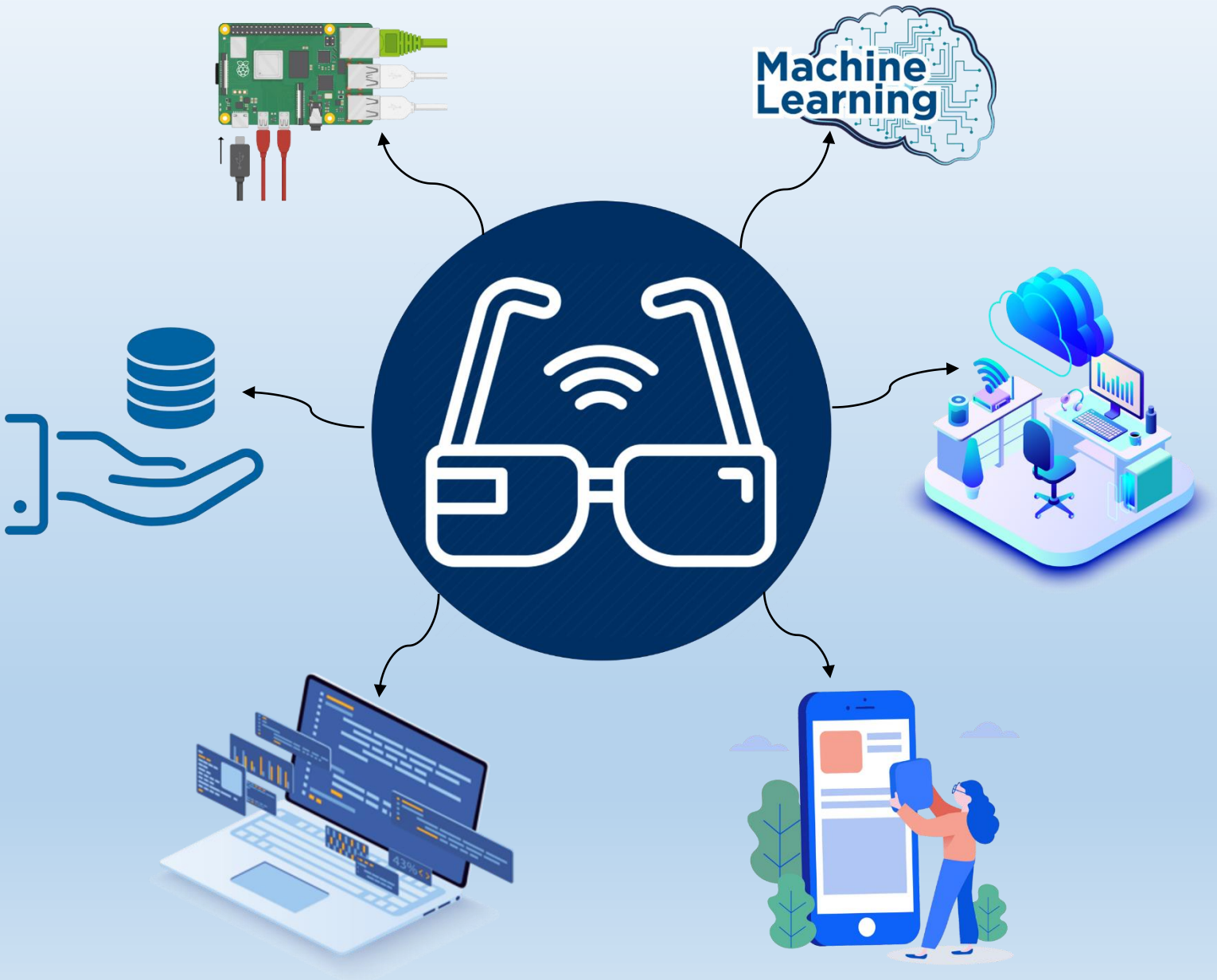
Supervisor: Dr/Khaled Saada

00

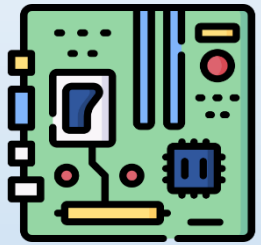
Intro

Ahmed Ghanem

Project Overview



Our Team



Mohamed Basiony
Hardware



Alaa Elmaghlany
UI-UX designer



Mohamed Beshara
Mobile Application



Sondos Elsafy
Mobile Application



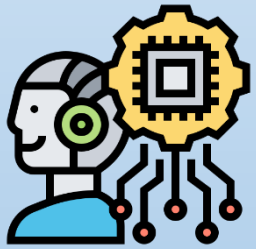
Mohamed El-Zarka
Front end



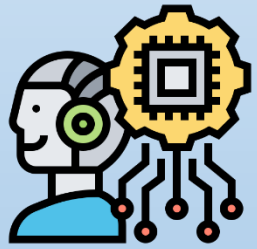
Manal said
Front end



Ahmed Ghanem
Cloud



Rahma Oshba
Machine Learning



Youssef Elsheheimy
Machine Learning



Ahmed Elneziliiy
Backend

01

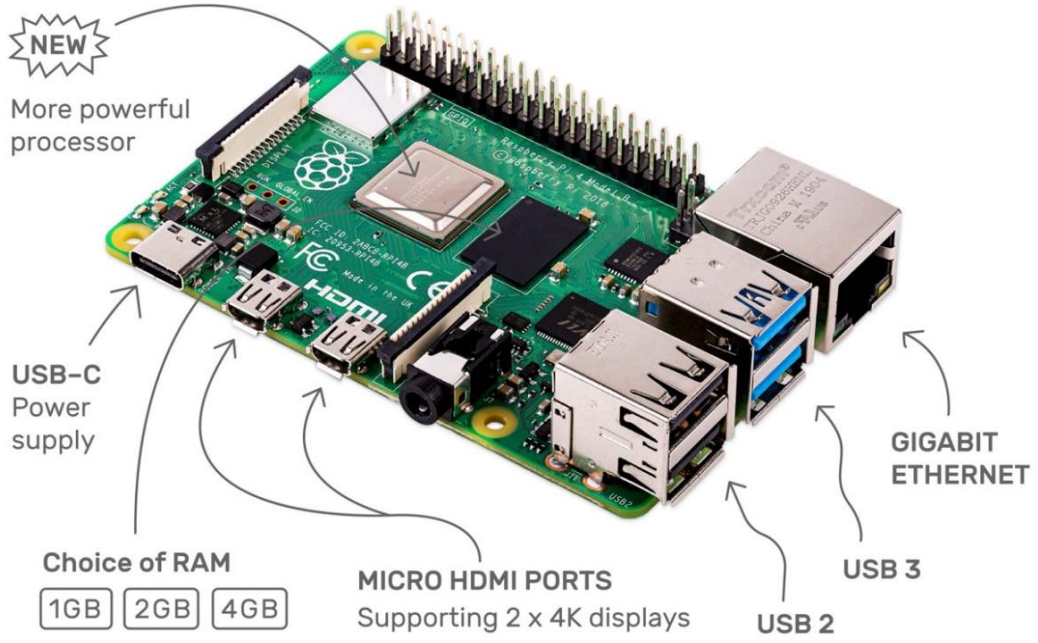
Hardware

Mohamed Basiony

1 Raspberry Pi

Why Do Some Engineers Choose RPi?

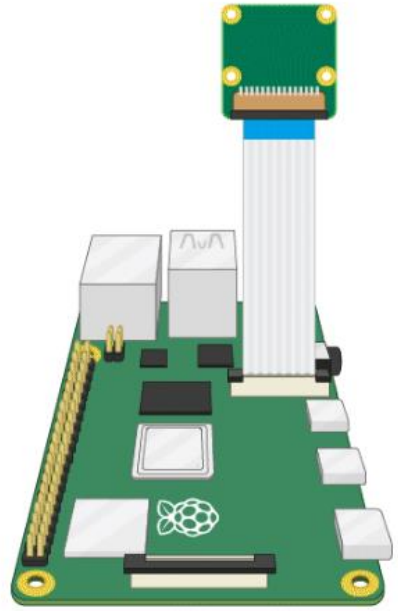
- Low cost
- Huge processing power in a compact board
- Many interfaces (HDMI, multiple USB, Ethernet, onboard Wi-Fi and Bluetooth, many GPIOs, USB powered, etc.)
- Supports Linux, Python (making it easy to build applications)
- Readily available examples with community support
- Developing such an embedded board is going to cost a lot of money and effort

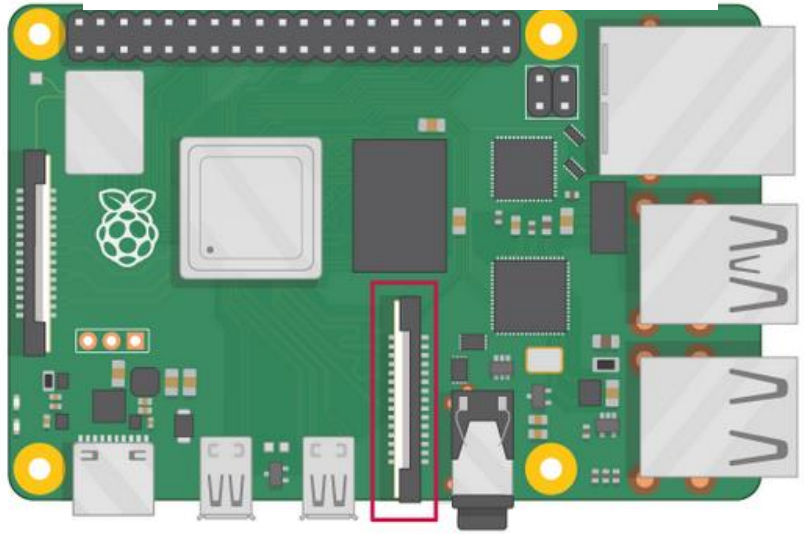


2 Camera

The V2 camera module supports 1080p30, 720p60 and VGA90 video modes.

The picamera library is well-written and powerful. It allows taking pictures, recording videos, streaming videos, streaming videos to a network, and even image processing with openCV.





3 Glasses design





[Smart Vision: all about the Envision (AI powered smart glass (amazon.com)]

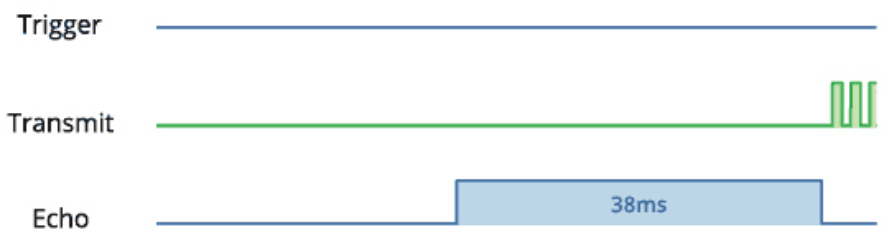
2 How Ultrasonic Work?

It all starts when the trigger pin is set HIGH for 10µs. In response, the sensor transmits an ultrasonic burst of eight pulses at 40 kHz. This 8-pulse pattern is specially designed so that the receiver can distinguish the transmitted pulses from ambient ultrasonic noise.

These 8 ultrasonic pulses travel through the air away from the transmitter. Meanwhile the echo pin goes HIGH to initiate the echo-back signal.



Last Minute ENGINEERS.com

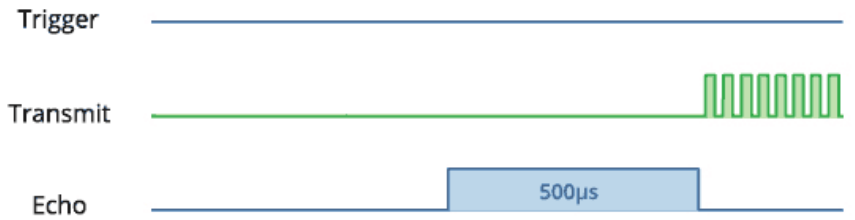


If those pulses are reflected back, the echo pin goes low as soon as the signal is received. This generates a pulse on the echo pin whose width varies from 150 µs to 25 ms depending on the time taken to receive the signal.





Last Minute ENGINEERS.com



3 Connect the ultrasonic to Raspberry :

VCC to Pin 2 (VCC)

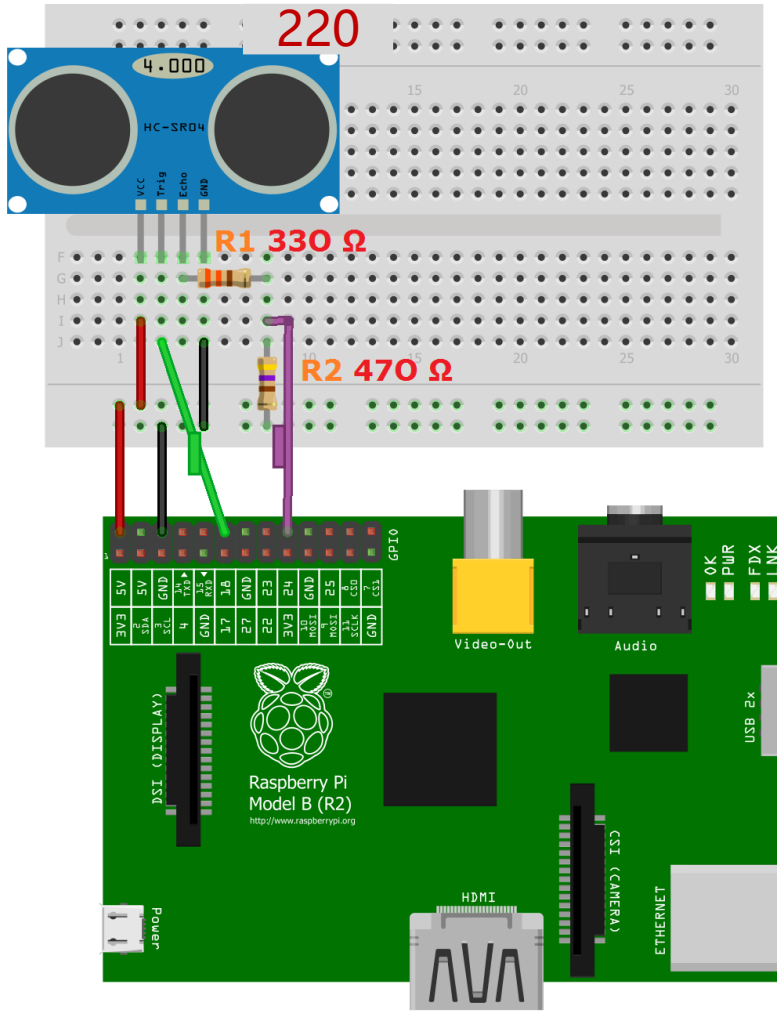
GND to Pin 6 (GND)

TRIG to Pin 11 (GPIO 17)

Connect ECHO to voltage divider to pin 13

(GPIO 27).

We do this because the GPIO pins only tolerate maximal 3.3V.



02

UI-UX

Alaa Elmaghlany

1 Why we use UI/UX?

Good design helps the developer understand the system

Saves time for developers

Early detection of problems, if any

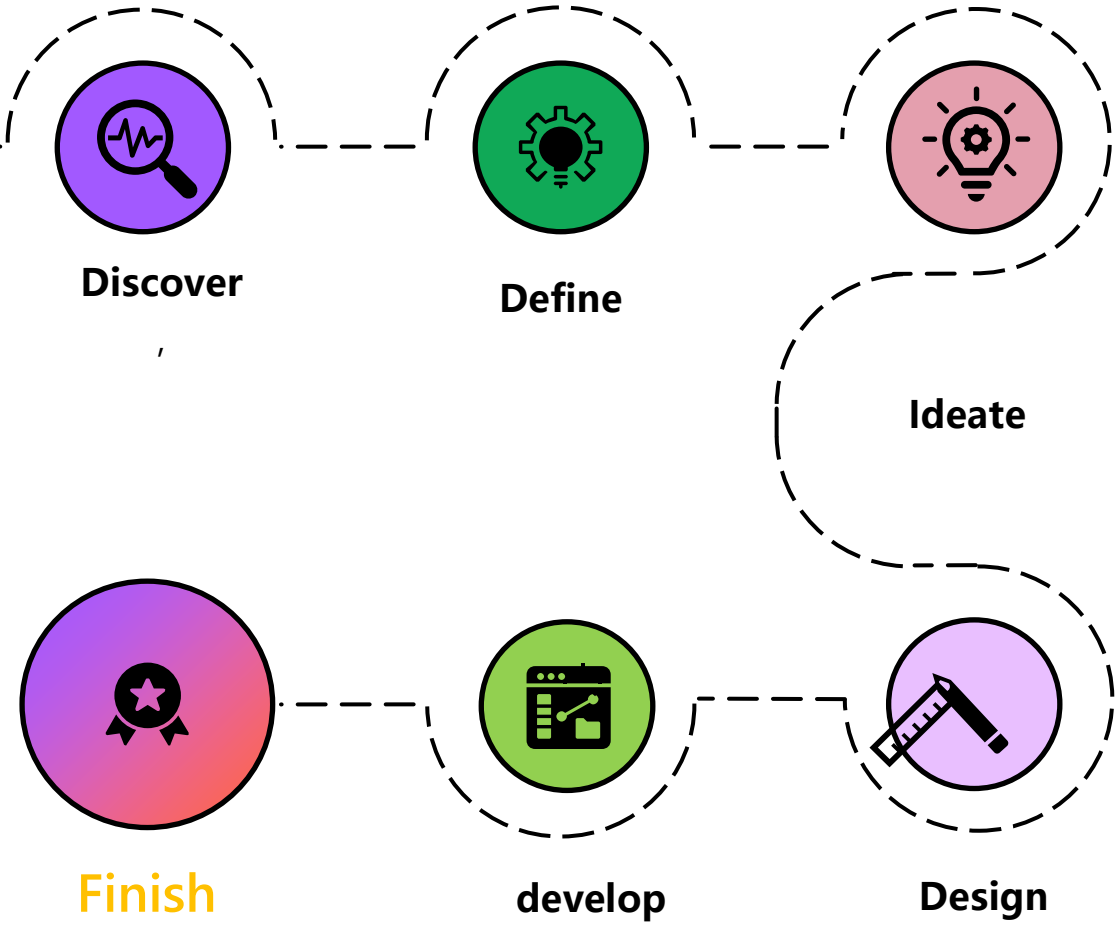
Easily modify the system



2 PROCESS

focus on the users and their needs in each phase of the design process





3 PROCESS

Define

I asked some questions that help us define the application more specifically

What does it do?

Connecting the glasses to the application and increasing the features to make the life of the blind easier

How does it work?

By use Talkback for Android
And Siri for iPhone

Who benefits from it ?

The blind and the facilities of the blind

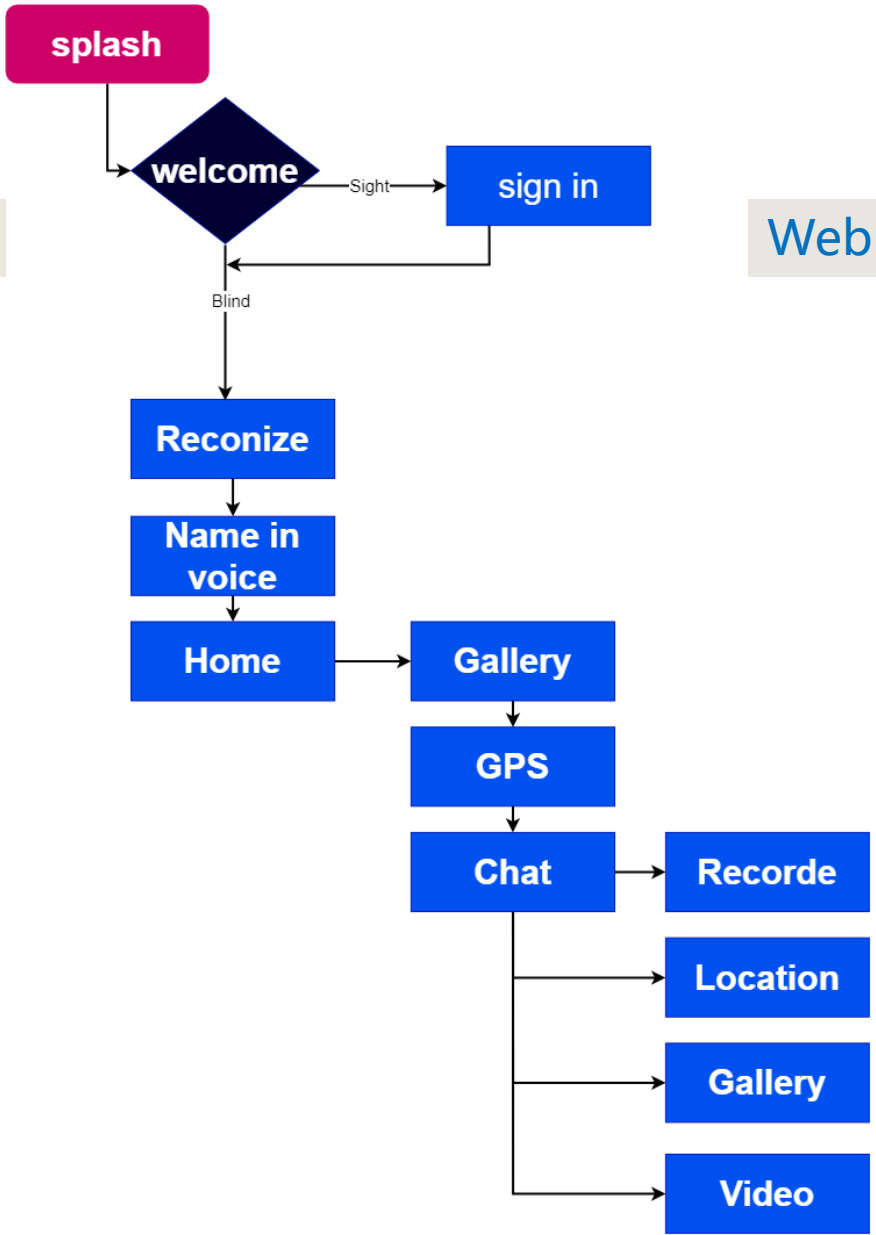
Why would someone use this?

He helps the blind people to follow him all the time and help the blind person to lead a better life

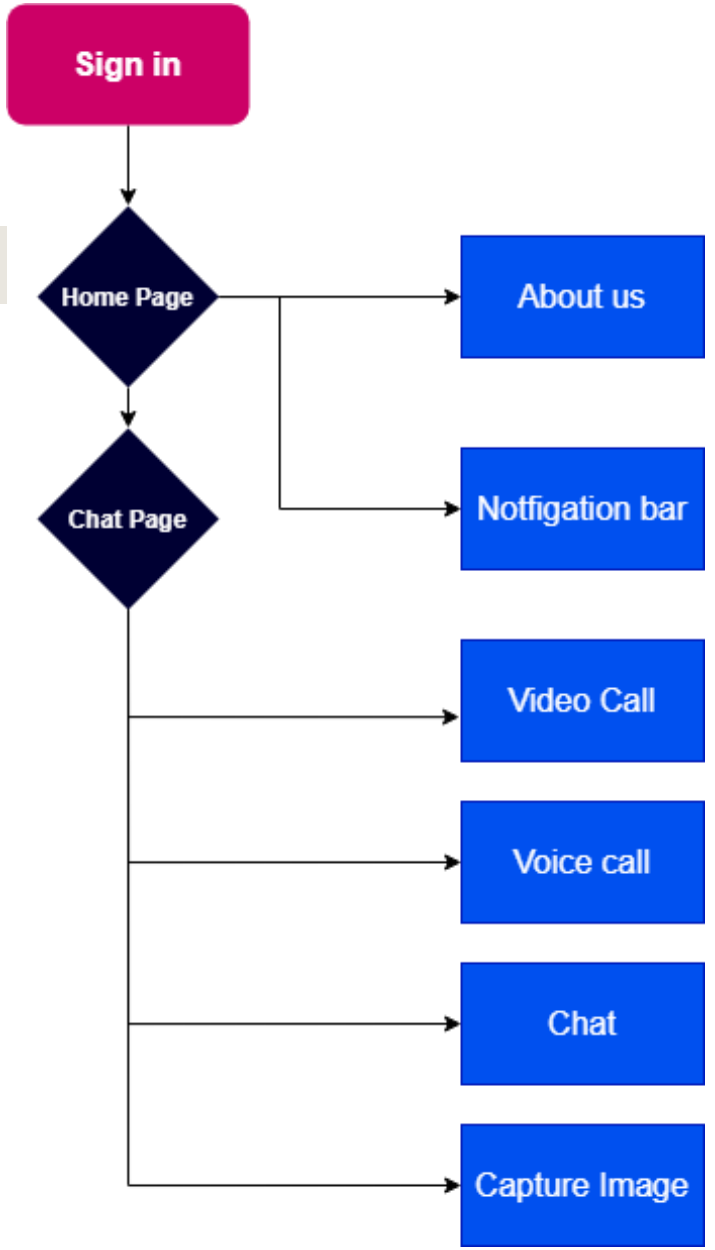
3 PROCESS

Ideate App

App Flow chart



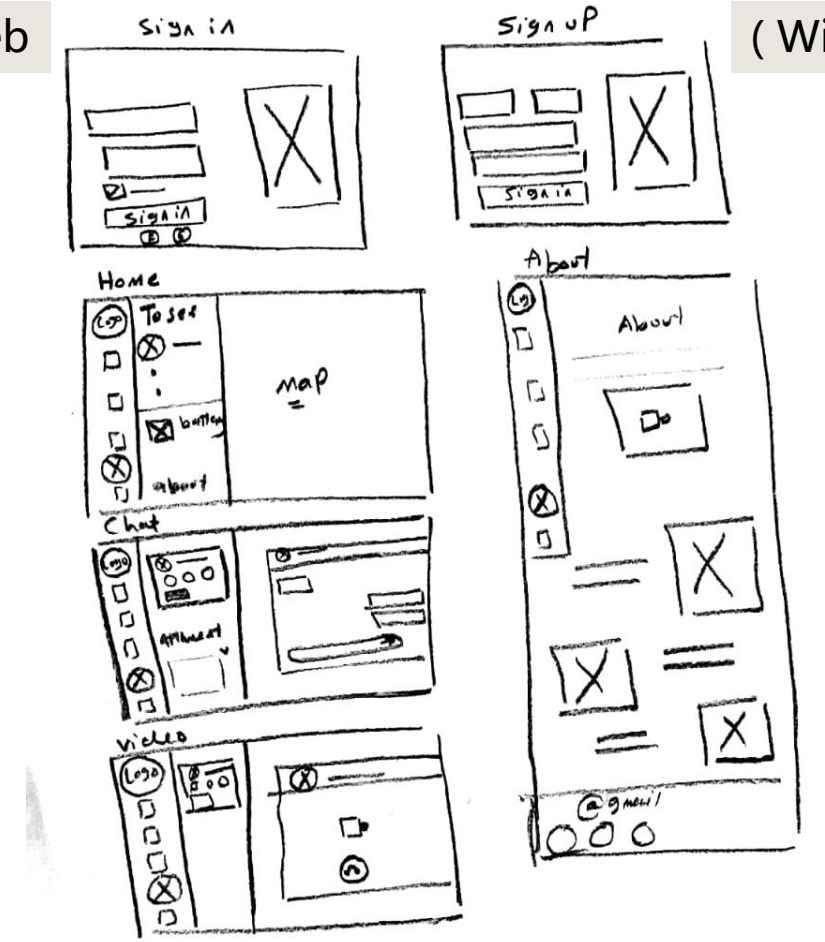
Web Flow chart



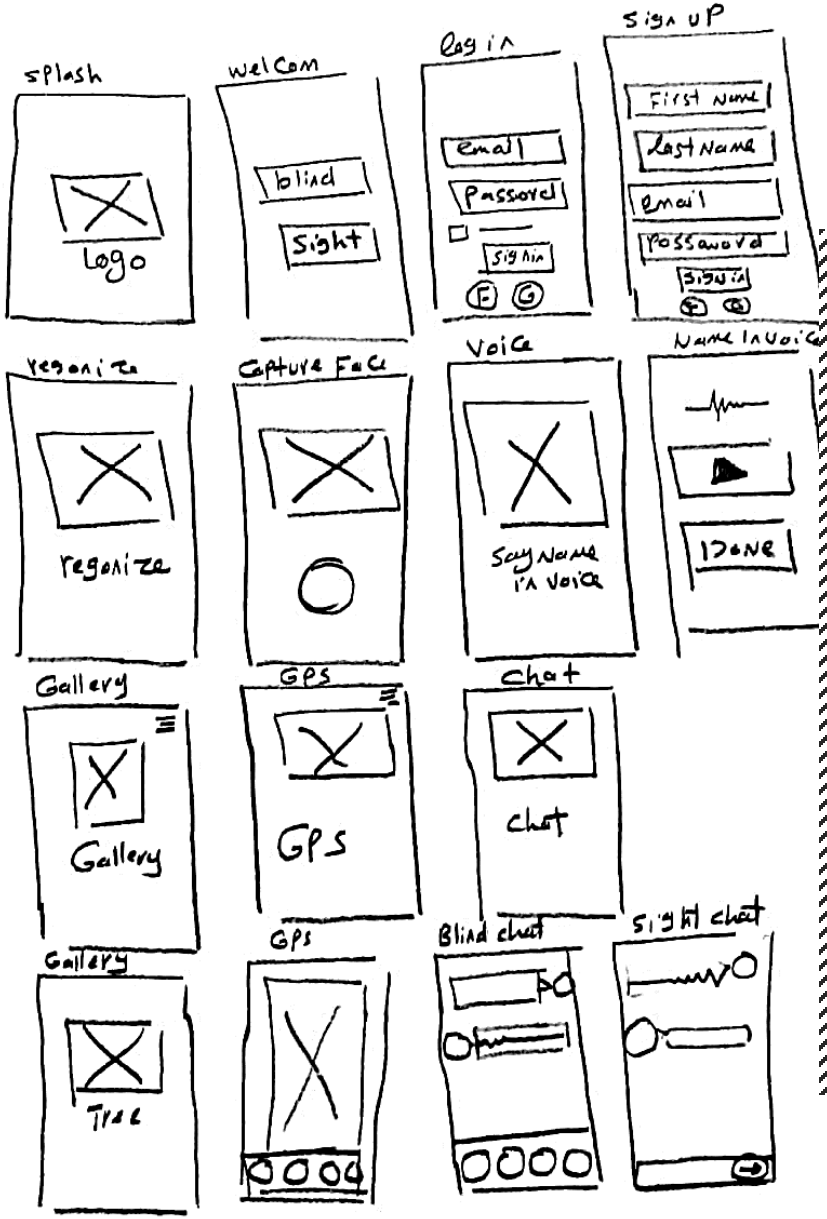
3 PROCESS

Design Low fidelity

(Wireframing) web



(Wireframing) App



03

Mobile Application

Mohamed Beshara
Sondos Alsafy

2 What is Flutter?

Flutter is an open source software development toolkit

Developed and maintained by Google

Allows cross platform development in one codebase

Flutter is built around the Dart programming language

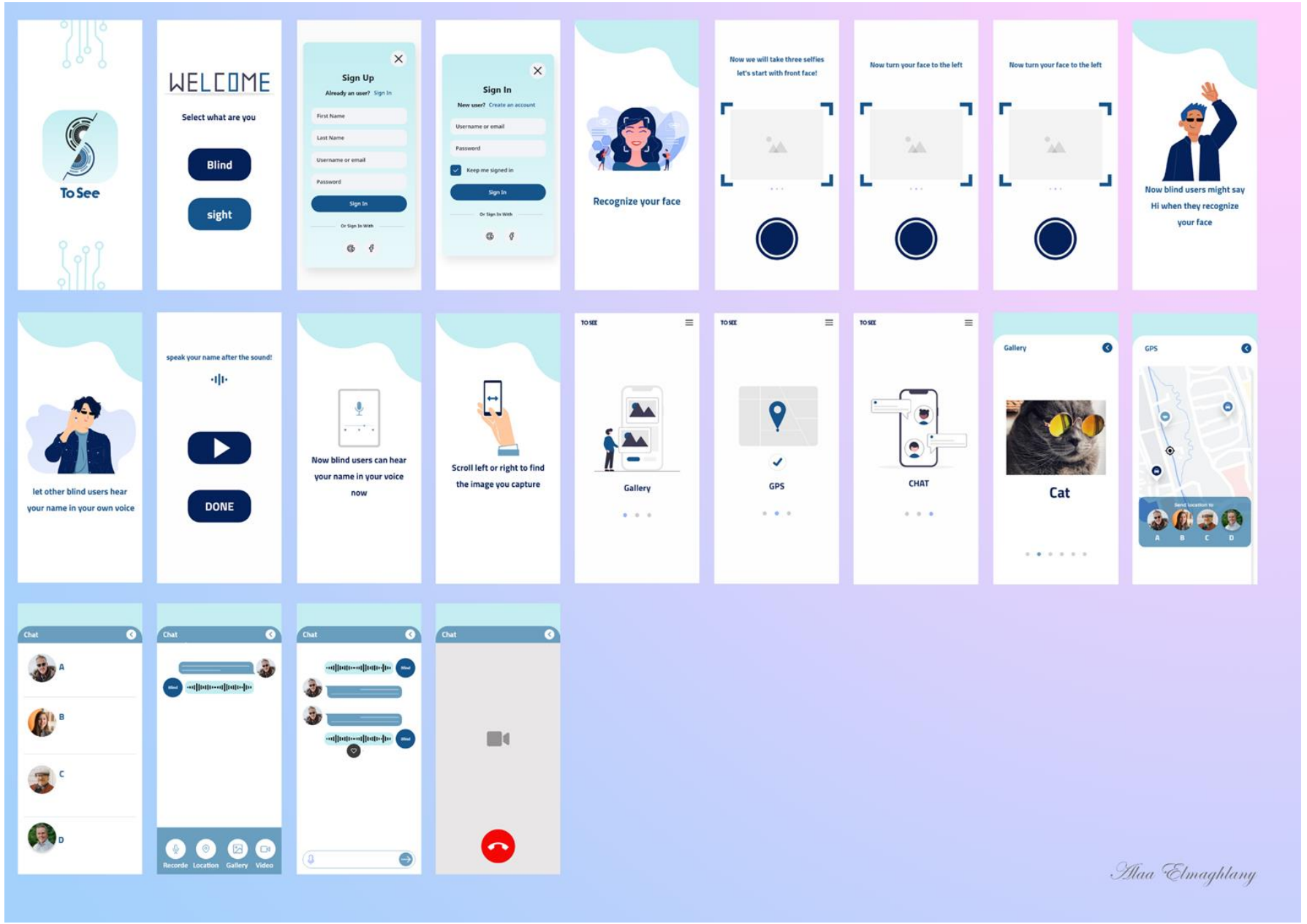


3 Advantages of flutter

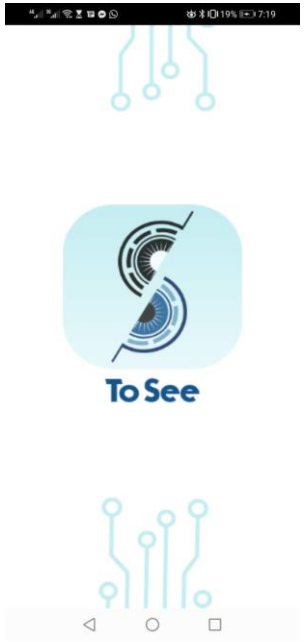


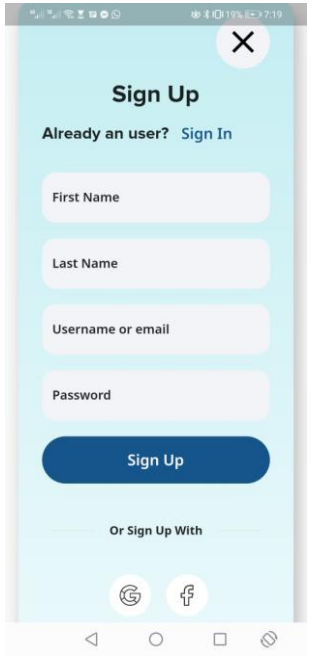
PROCESS

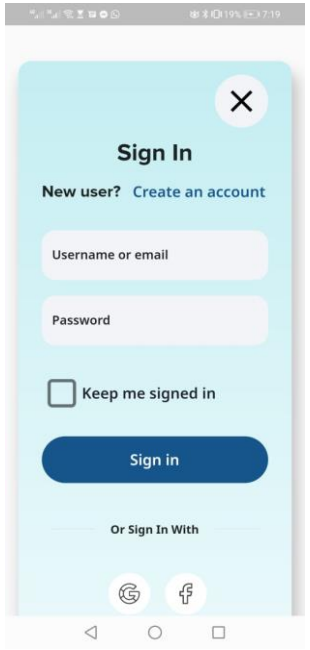
Design High fidelity



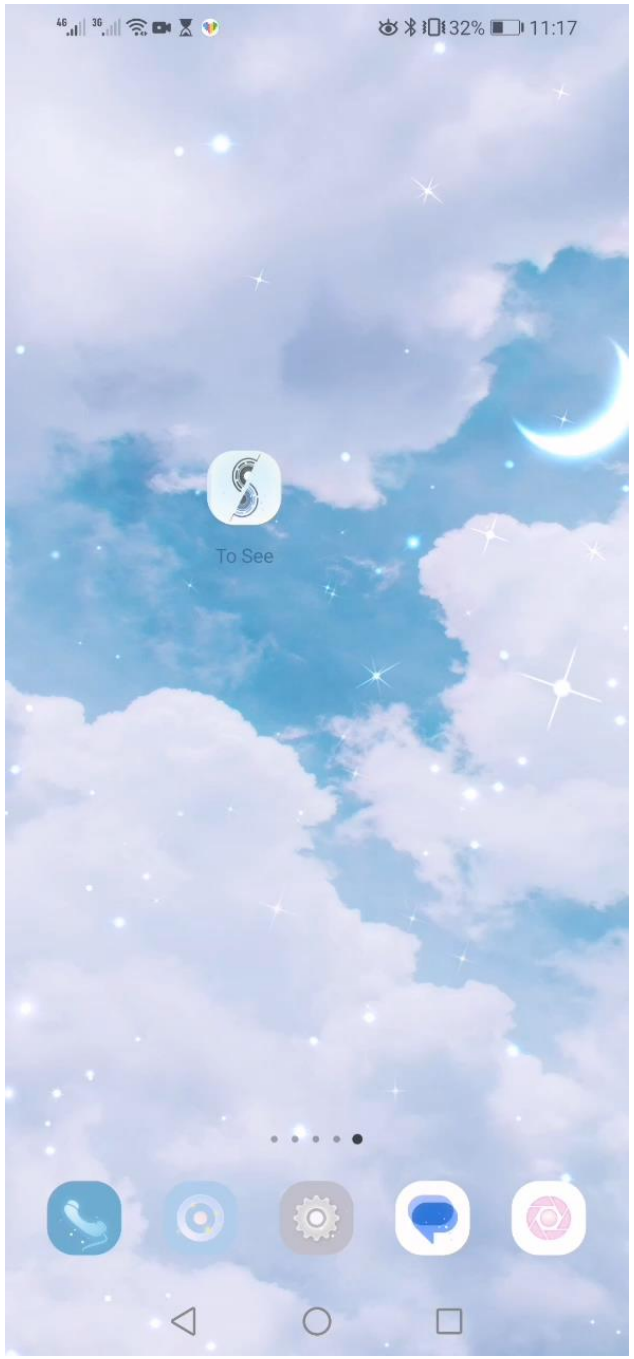
4 What we achieved..











04

Frond end developer

Mohamed El-zarka
Manal said

1 Frontend Frameworks & Libraries

React and Bootstrap



We are going to use **React.js** along side with **Bootstrap5** for styling in order to implement the client side of the web app.

React: Provides the modularity of a single page application and the overall better performance.



Bootstrap5: makes it easy for responsive design (good for phone or PC usage).



2 Design (From UI/UX design)

Low fidelity (Wireframing) app

React Components in our project

1. sign in
2. create
3. sidebar

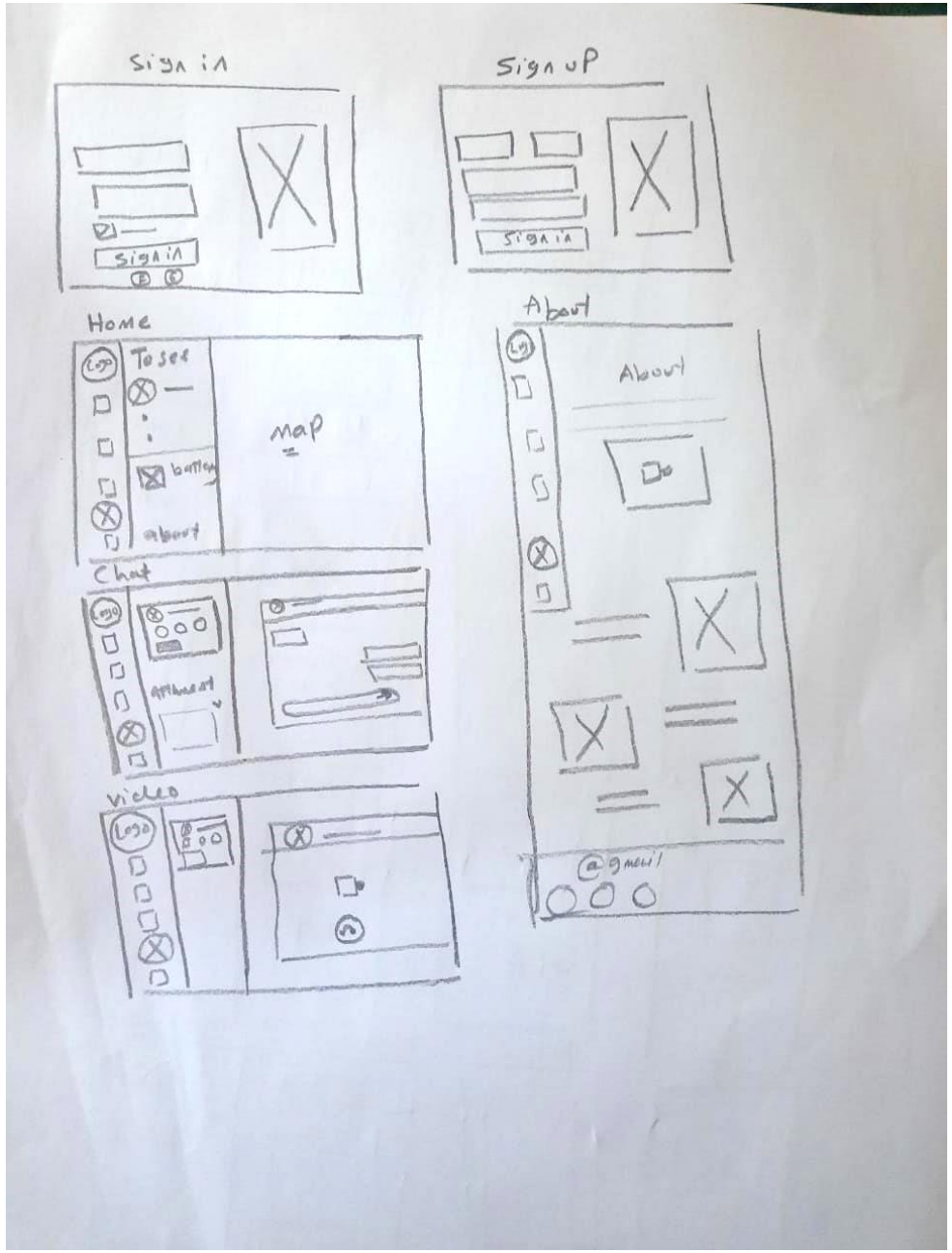
info

map

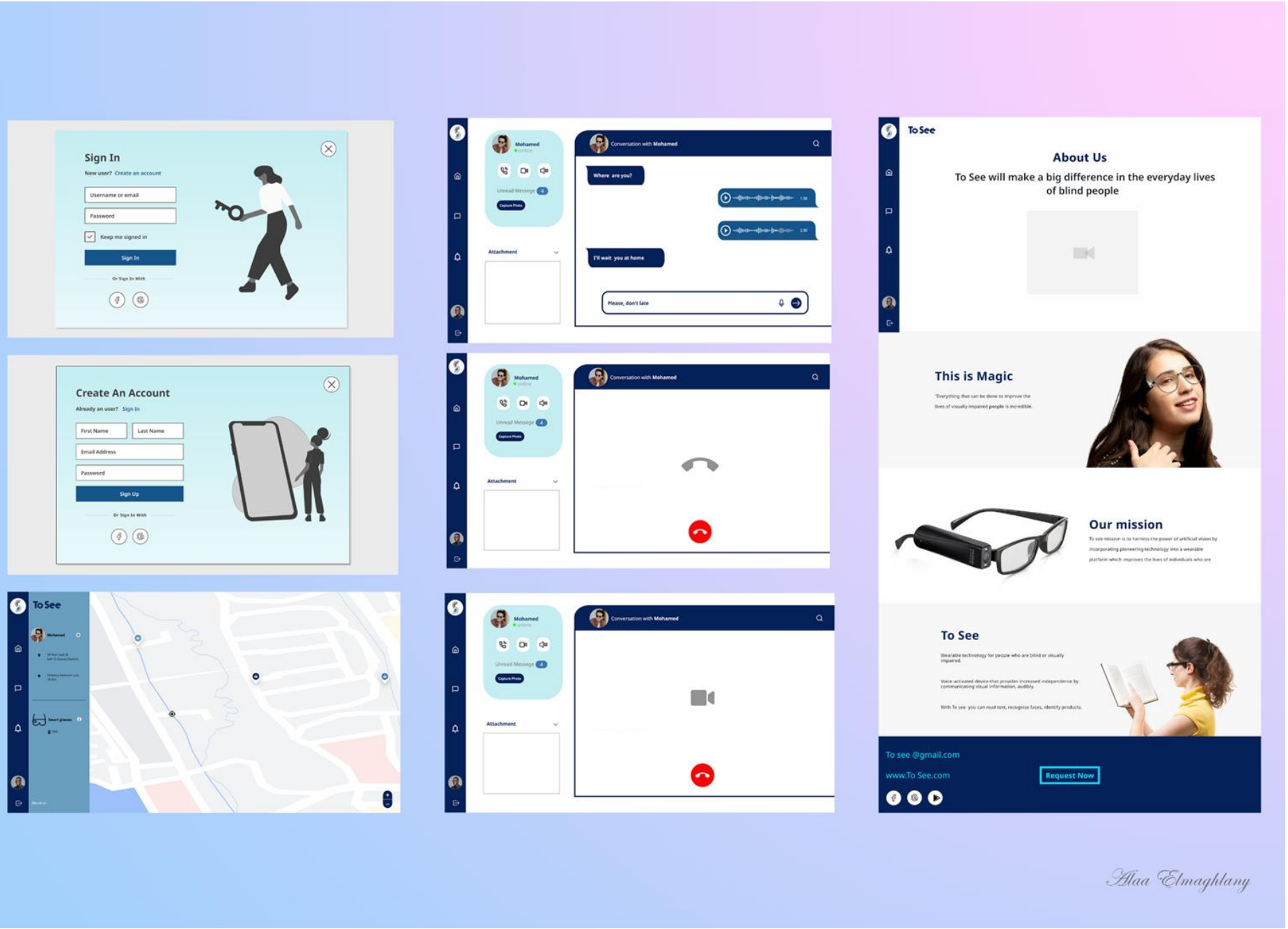
4. about (includes video)

footer

5. chat (messages List + input field)
6. WebRTC (VIDEO CALL)
7. Search results
8. WebRTC (AUDIO CALL)
9. gallery
10. notifications

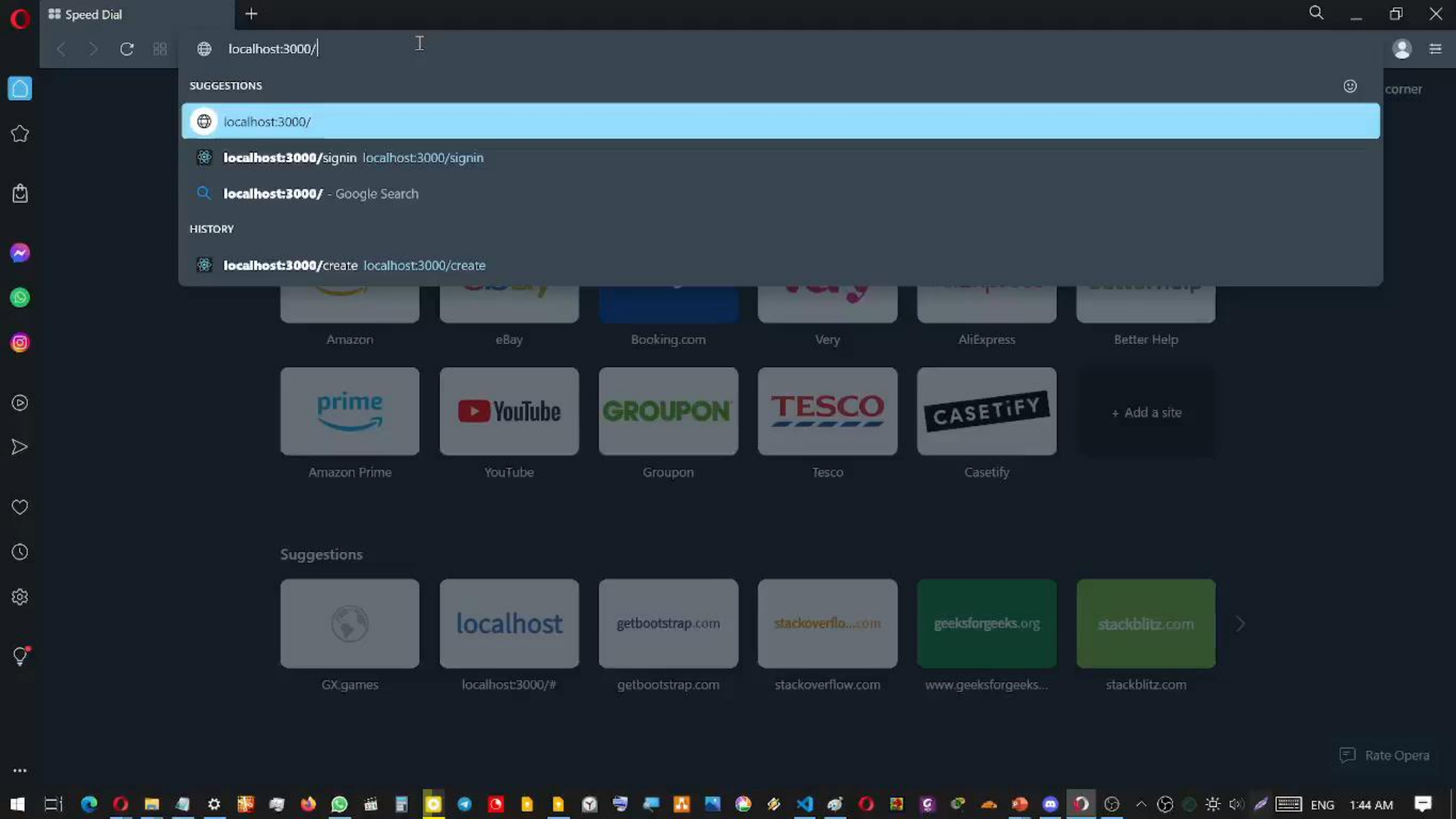


High fidelity (Wireframing) app



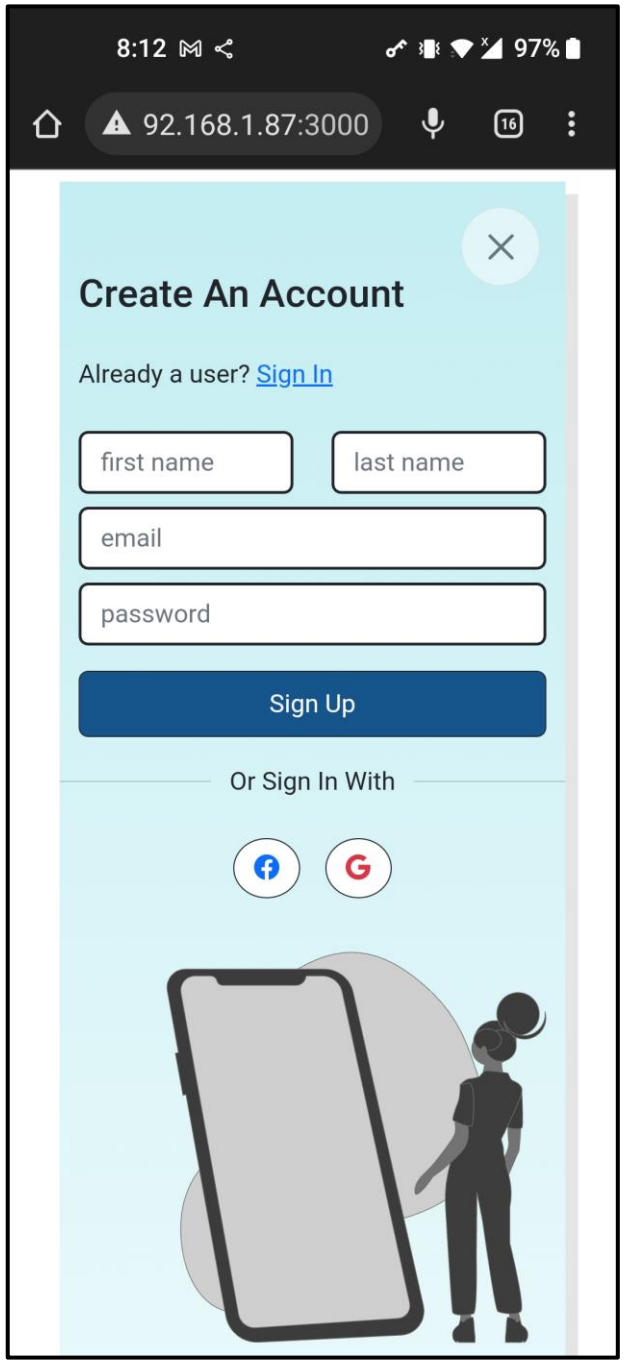
Alaa Elmaghlany

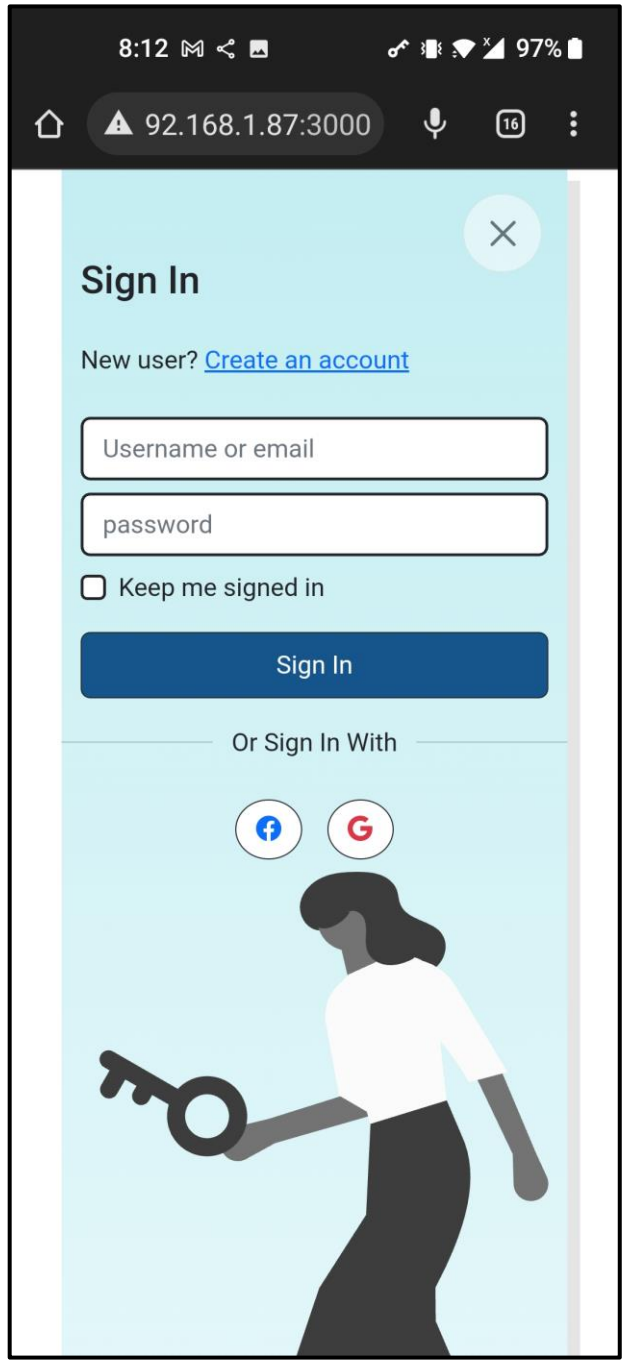
3 React App Demo preview


Speed Dial
localhost:3000/
SUGGESTIONS
localhost:3000/
localhost:3000/signin localhost:3000/signin
localhost:3000/ - Google Search
HISTORY
localhost:3000/create localhost:3000/create
Amazon
eBay
Booking.com
Very
AliExpress
Better Help
Amazon Prime
YouTube
Groupon
Tesco
Casetify
+ Add a site
Suggestions
GX.games
localhost:3000/#
getbootstrap.com
stackoverflow.com
www.geeksforgeeks...
stackblitz.com
Rate Opera
ENG
1:44 AM

4 Responsive design

Demo preview





05

Introduction to Cloud

Ahmed Ghanem

1 What is Cloud Computing ?

Cloud computing is a general term for anything that involves delivering hosted services over the internet.

like , applications, servers, data storage, development tools, networking capabilities, and more , managed by a cloud services provider (or CSP).

The CSP makes these resources available for a monthly subscription fee or bills them according to usage.



2 Types of Cloud computing



01

Infrastructure as a Service (IaaS)

IaaS contains the basic building blocks for cloud IT.



02

Software as a Service (SaaS)

SaaS provides you with a complete product that is run and managed by the service provider.



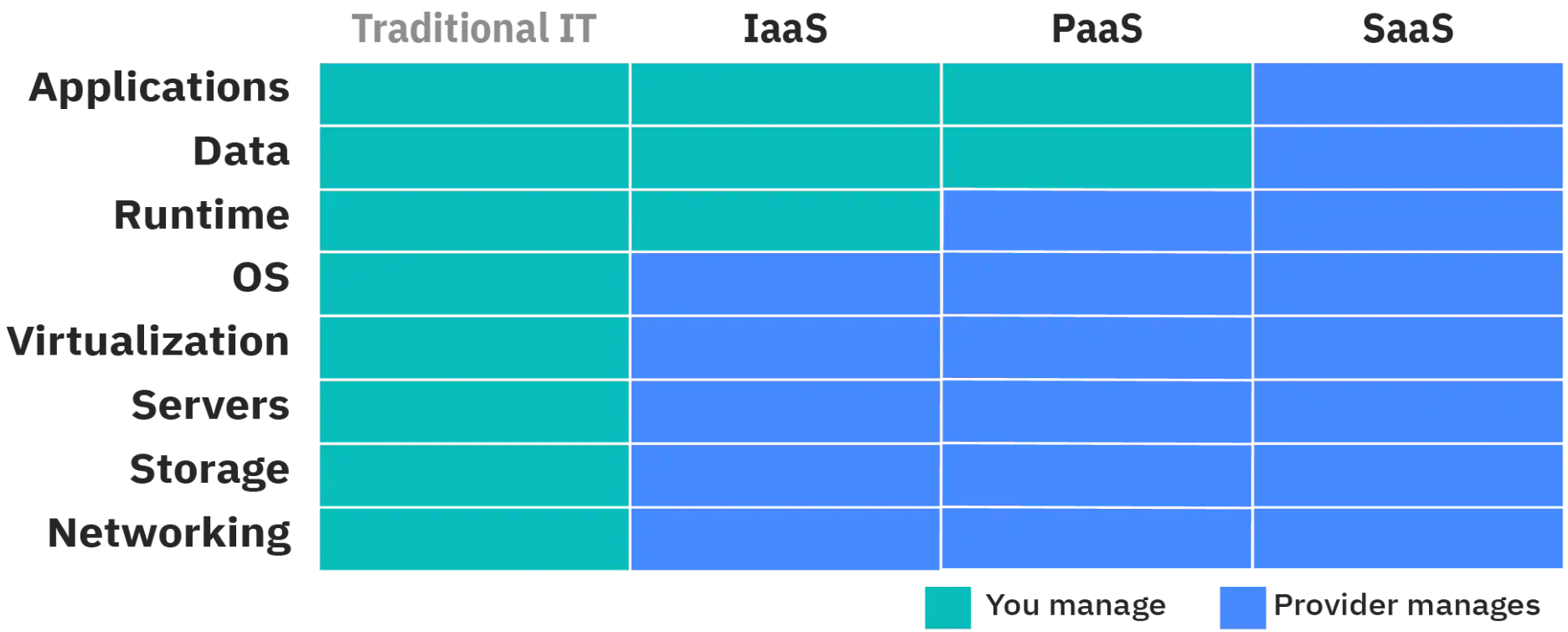
03

Platform as a Service (PaaS)

PaaS allows you to focus on the deployment and management of your applications.

3 Types of Cloud computing

Cloud Computing Services: Who Manages What?



4 Cloud use cases

Anything that involves storing and processing huge volumes of data at high speeds—and requires more storage and computing capacity than most organizations can or want to purchase and deploy on-premises—is a target for cloud computing. Examples include:

- Big data analytics
- Internet of Things (IoT)
- Artificial intelligence—particularly machine learning and deep learning applications



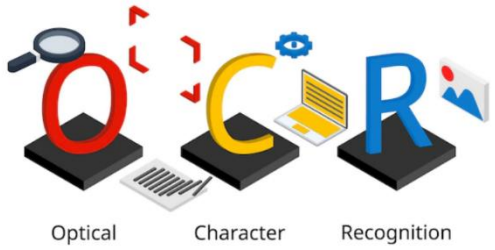
06

Machine Learning

Rahma Oshba
Youssef Elsheheimy

1 OCR "Optical character recognition"

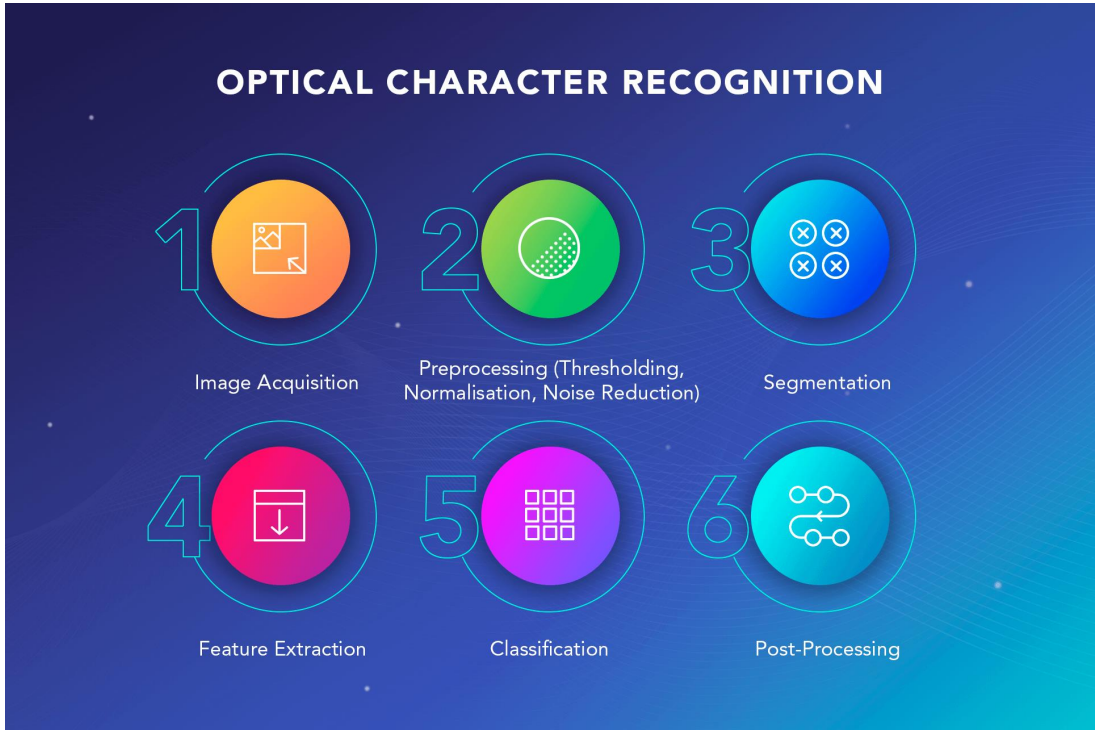
OCR analysis takes the input as a printed or Handwritten Digital Image And Converts It To a digital text format.



OCR has Two Components:

The First Part Is The Detection Of Text,
The Second Part Of OCR, Text Recognition

-Phase 1: Pre-processing
-Phase 2: Character recognition
-Phase 3: Post-processing



1 OCR

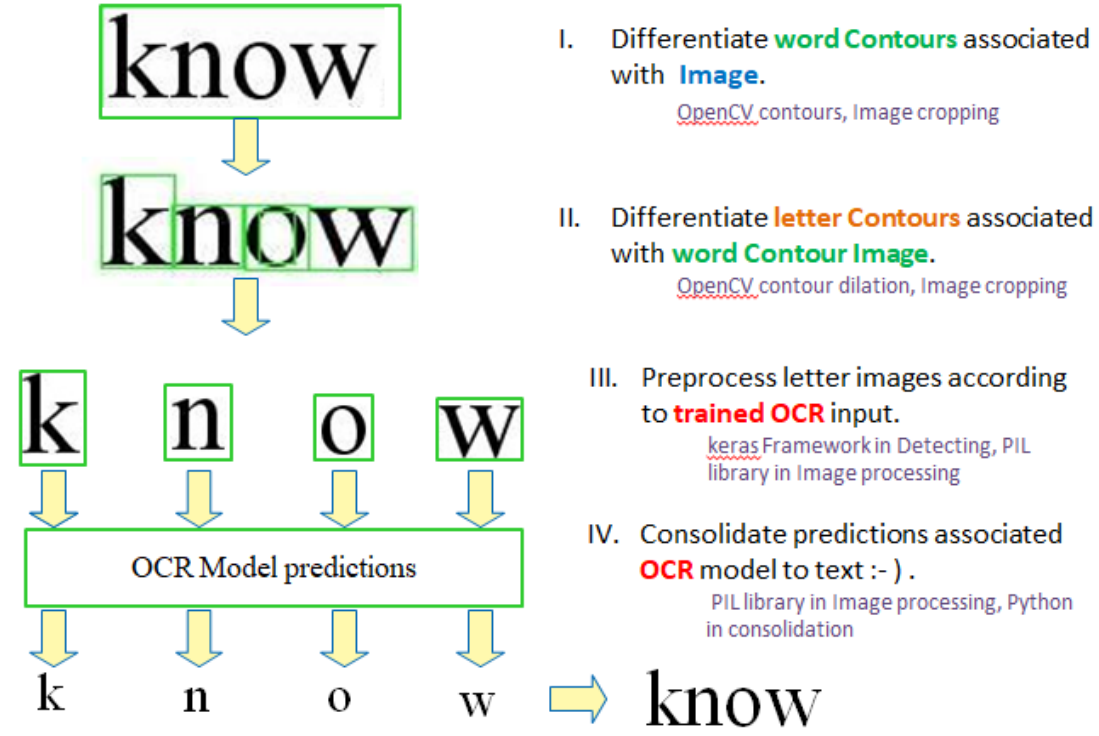
Applications of optical character recognition

Text-to-speech(Main)
Number-plate recognition
Banking and finances
Preservation of documents

OCR for blind people

For people with physical impairments such as blindness, OCR systems are a life-changing aspec of their daily lives.
optical character recognition technology is more enhanced than ever before by expanding to support hundreds of languages and dialects.

Optical Character Recognition flow diagram



2 Models







2 Models

Tesseract vs EasyOCR:

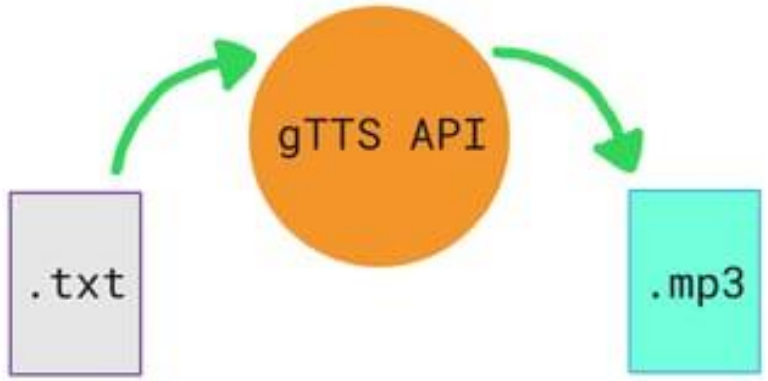
Tesseract is preferable for CPU wheras EasyOCR for GPU machine.

Tesseract works better on character level, while EasyOCR does a better job on words.

gTTS (Google Text-to-Speech):

The gTTS module can be used extensively on other languages

Python library and CLI tool to interface with Google Translate text-to-speech API. We will import the gTTS library from the gtts module which can be used for speech translation.



07

Back end developer

Ahmed Elneziliiy

1 Application Programming Interface

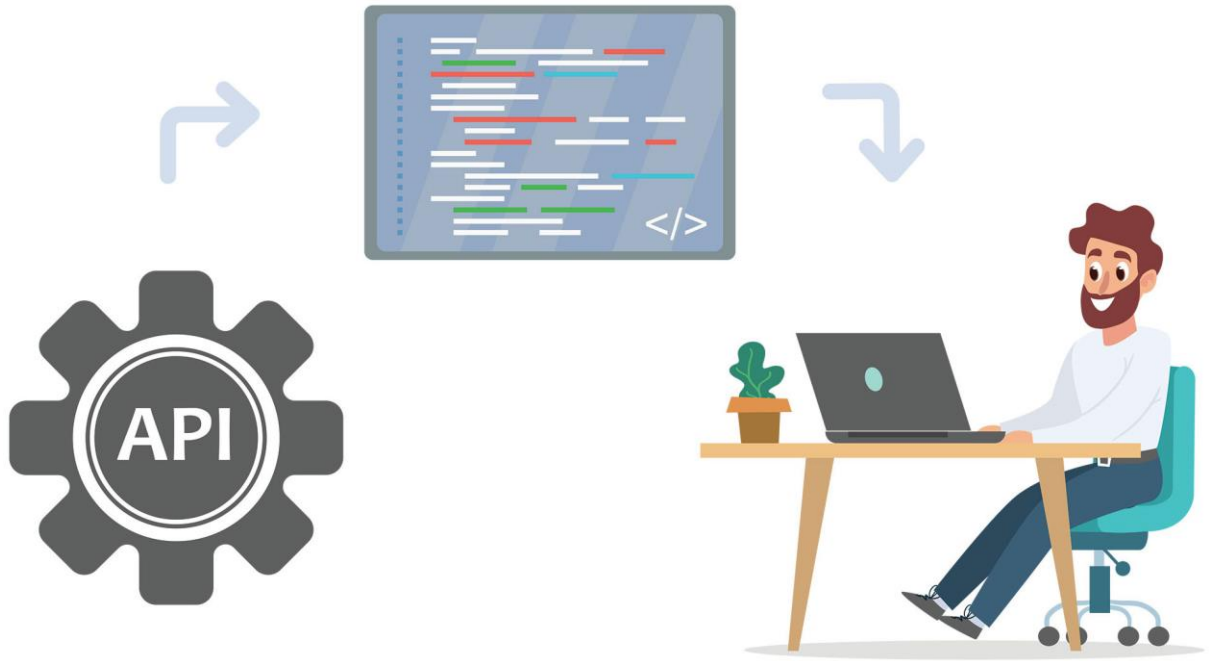
Definition

API stands for Application Programming Interface.

It is a collection of functions, procedures, or methods that are available to be executed by other software applications.

Why we use it ?

Its main purpose is to offer access to certain services and provide communication between software components .



2 RESTful API

Overview

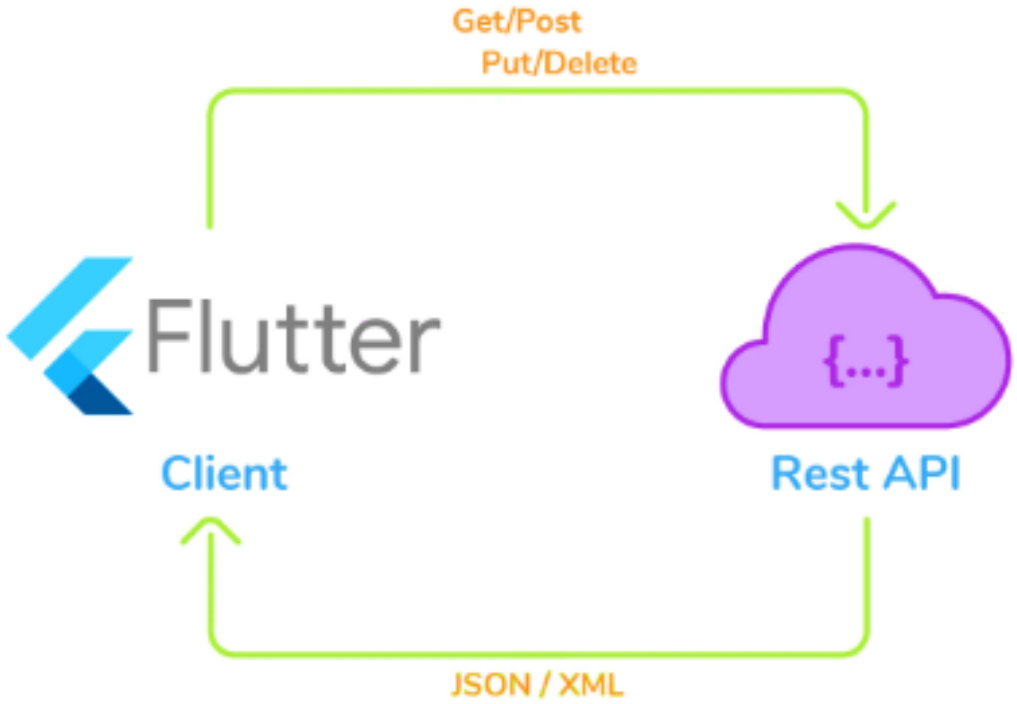
REST stands for(REpresentational State Transfer)

- REST is a software architectural style that defines the set of rules to be used for creating web services.

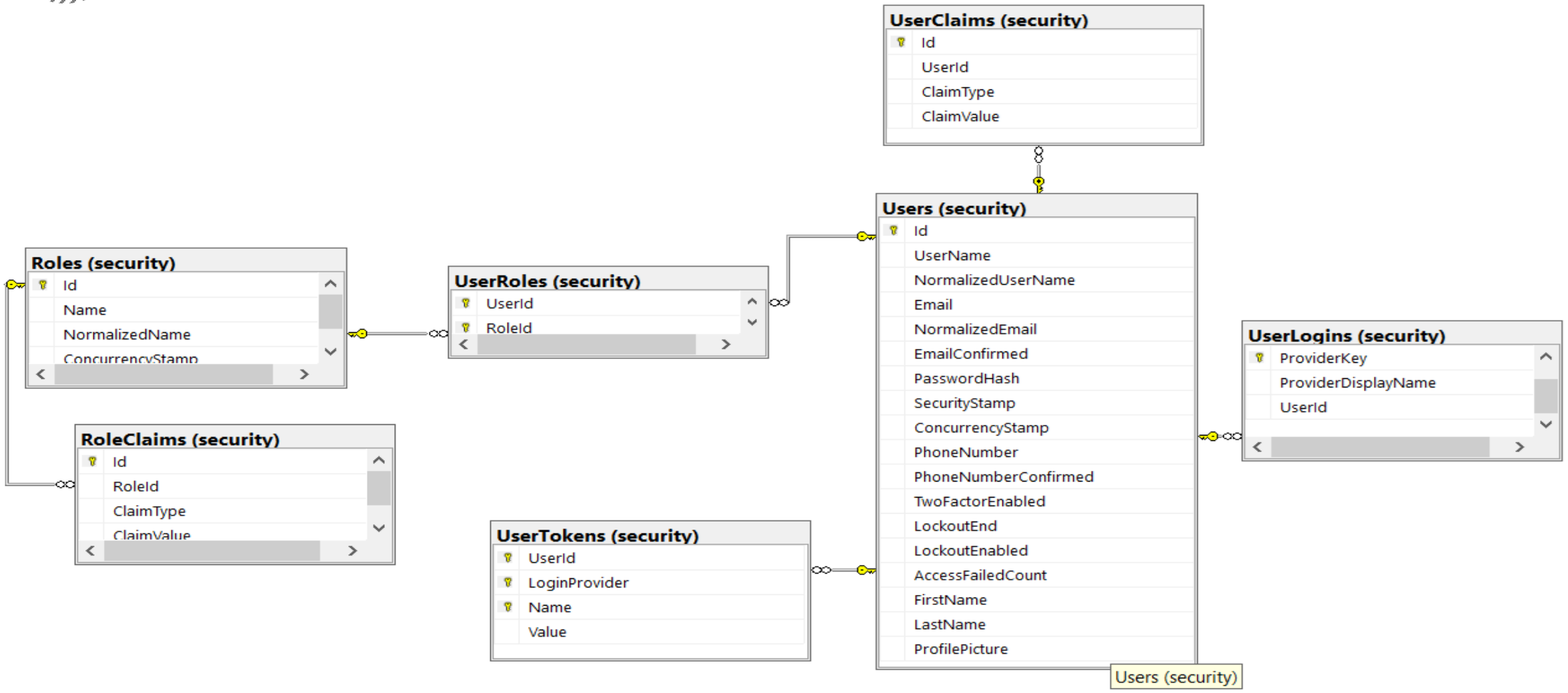
Web services which follow the REST architectural style are known as RESTful web services.

using a systems happen through Internet’ s Hypertext Transfer Protocol (**HTTP**)

Roy Fielding



3 Database Schema:-



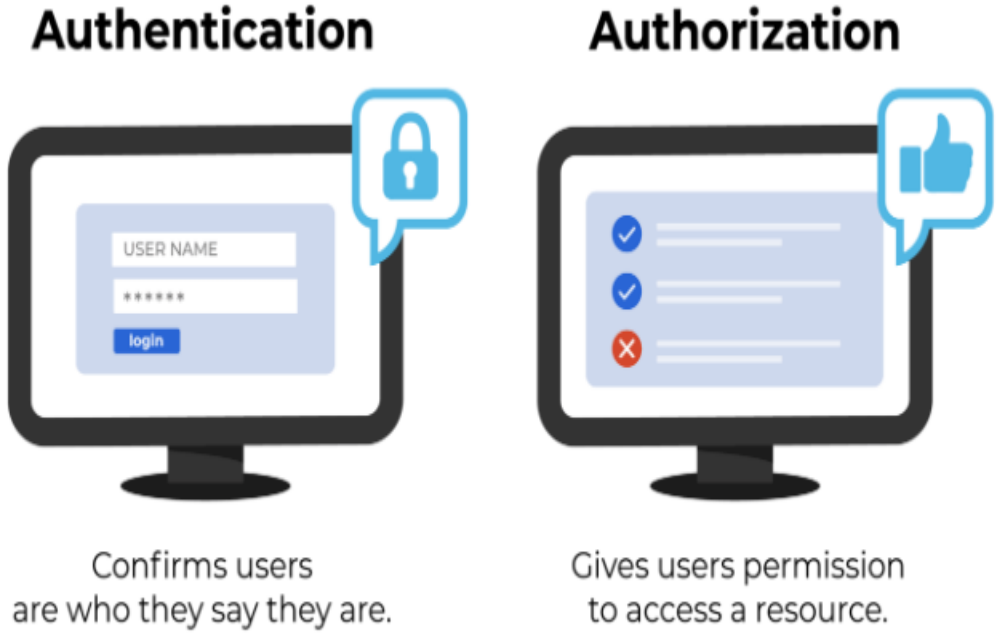
4 Authentication And Authorization

What Is Authentication?

Authentication is the act of validating that users are whom they claim to be. This is the first

What Is Authorization?

Authorization in system security is the process of giving the user permission to access a specific resource or function.



Authentication and authorization are two vital information security processes that administrators use to protect systems and information. Authentication verifies the identity of a user or service, and authorization determines their access rights.

Microsoft Identity framework .

Thank you